



Microcontroller & Interfacing

CE205T

	CLO-2		CLO-3					Total
Part	B	C	A	D	E	F	G	
Marks	50	50	50	50	50	50	50	
Obt.								

Microcontroller & Interfacing - CE205T

Environmental Monitoring and Warning System

Group Number: 8

1. MUHAMMAD MUNEEB JAVED - BSCE23017
2. HASSAN SUBHANI - BSCE23034
3. TAIMOOR SHAHZAMAN - BSCE23046

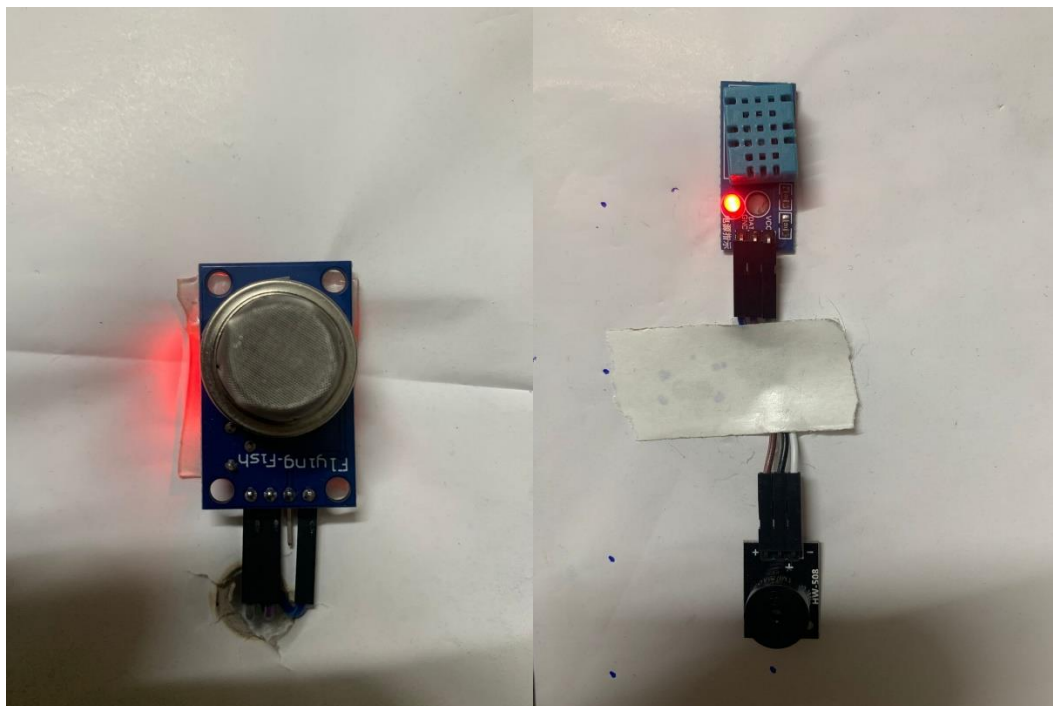
A. Overview [CLO-3, 50 Marks]

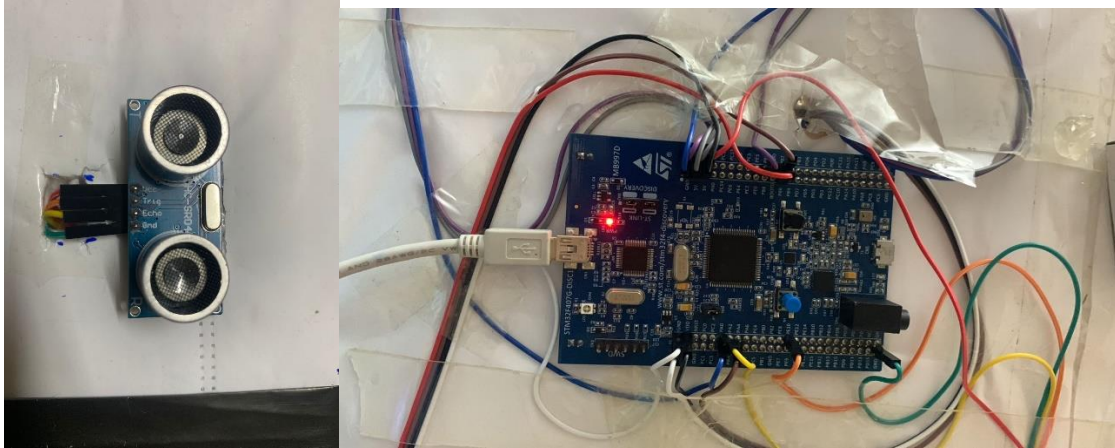
This project implements an environmental monitoring and warning system using the STM32F4 microcontroller platform. The system continuously monitors multiple environmental parameters including temperature, humidity, air quality, and distance to nearby objects. When environmental conditions exceed safe thresholds, the system activates visual and audio warnings to alert users.

The system features real-time data acquisition from multiple sensors, data processing algorithms to convert sensor readings into meaningful metrics, and a user-friendly OLED display interface to present the information. The system has both monitoring and warning functionalities, making it suitable for applications like indoor air quality monitoring, laboratory safety, or industrial environment monitoring.

GOALS

- Create a comprehensive environmental monitoring system using multiple sensors with different communication protocols
- Implement real-time data acquisition, processing, and display on an OLED screen
- Develop a warning system that activates when environmental conditions become hazardous
- Demonstrate proper utilization of multiple STM32 peripherals (Timers, ADC, I2C, Interrupts)





B. List of Components Used [CLO-2, 50 Marks]

Component	Description	Operating Principle	Cost (Est. Rs)	Datasheet Link
STM32F407VG	32-bit ARM Cortex-M4 microcontroller	168MHz CPU with FPU, multiple peripherals	ISSUED	STM32F407VG Datasheet
DHT11	Temperature and humidity sensor	Capacitive humidity sensor and thermistor	400	DHT11 Datasheet
MQ135	Air quality sensor	Electrochemical gas sensor that changes resistance based on gas concentration	500	MQ135 Datasheet
HC-SR04	Ultrasonic distance sensor	Uses ultrasonic pulses to measure distance by timing echo	350	HC-SR04 Datasheet
SSD1306	128x64 OLED display	Monochrome OLED display with I2C interface	600	SSD1306 Datasheet
LED	Warning indicator	Light emitting diode for visual warnings	STM32	Generic
Buzzer	Audio warning device	Piezoelectric element that produces sound	200	Generic

Sensor Operating Principles:

1. DHT11 Sensor:

- Uses a capacitive humidity sensing element and thermistor for temperature
- Outputs digital signal using a single-wire serial interface
- Temperature Range: 0-50°C, Humidity Range: 20-90% RH
- Requires precise timing for communication protocol

2. MQ135 Gas Sensor:

- Semiconductor-based gas sensor that changes resistance when exposed to gases
- Contains a heating element to activate the sensing element
- Sensitive to NH₃, NO_x, alcohol, benzene, smoke, CO₂, etc.
- Output is an analog voltage that relates to gas concentration

3. HC-SR04 Ultrasonic Sensor:

- Uses ultrasonic pulses (40kHz) to measure distance
- Transmits a short pulse, then measures the time until echo returns
- Uses the speed of sound in air to calculate distance
- Range: 2cm to 400cm with 3mm resolution

C. Peripherals of STM Microcontroller being used [CLO-2, 50 Marks]

Peripheral	Configuration	Purpose
GPIO	Multiple pins configured as input/output	DHT11 communication, HC-SR04 trigger, LED/buzzer control
Timer 1 (TIM1)	1MHz clock (72MHz/72), counter mode	Precise microsecond delays for sensor timing requirements
Timer 2 (TIM2)	1MHz clock, input capture mode	HC-SR04 echo pulse duration measurement
ADC1	12-bit resolution, channel 1	Reading analog values from MQ135 air quality sensor
I2C1	400kHz clock, 7-bit addressing	Communication with SSD1306 OLED display
Interrupt Controller	Timer input capture interrupts	Processing HC-SR04 echo signal timing
RCC	72MHz system clock	System clock configuration

Peripheral Utilization Details:

1. **Timer 1 (TIM1):**

- Configured with 1 μ s resolution for precise timing control
- Used for generating exact microsecond delays required by sensor protocols
- Essential for DHT11 sensor's bit-banged communication protocol

2. **Timer 2 (TIM2):**

- Configured in input capture mode with interrupts
- Captures rising and falling edges of HC-SR04 echo signal
- Measures time difference to calculate distance with high precision

3. **ADC1:**

- Configured for 12-bit conversion (0-4095 value range)
- Samples MQ135 sensor output voltage
- Converted to PPM and AQI through calibration formulas

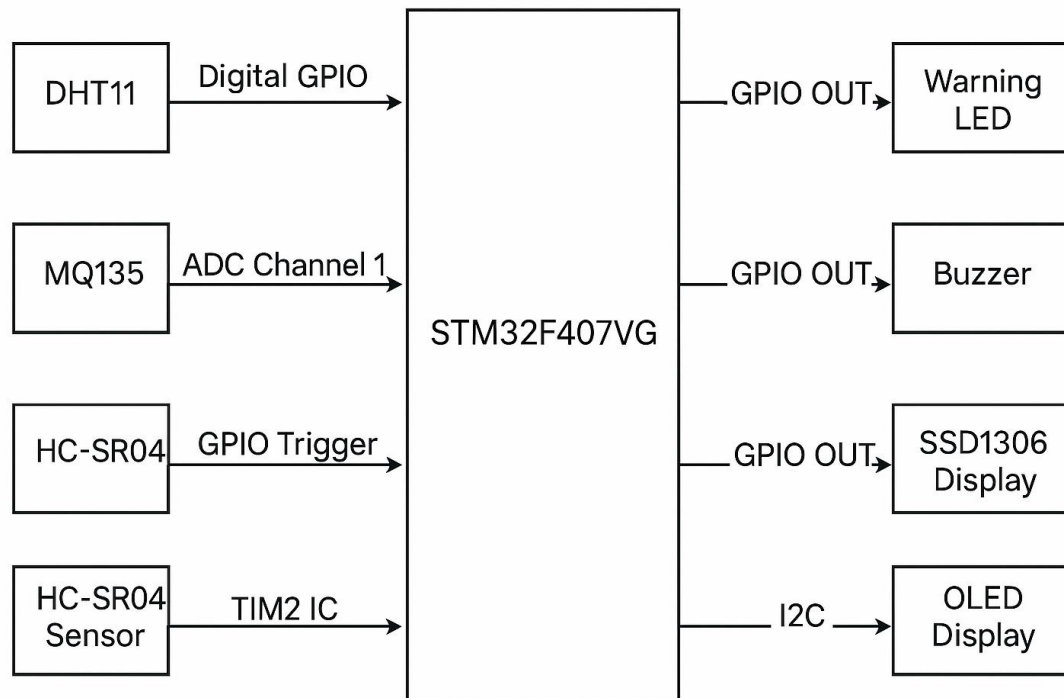
4. **I2C1:**

- Configured in fast mode (400kHz)
- Handles all communication with SSD1306 OLED display
- Manages display initialization, updates, and rendering

5. **Interrupt System:**

- Input capture interrupts for precise distance measurement
- Priority-based handling for real-time responsiveness

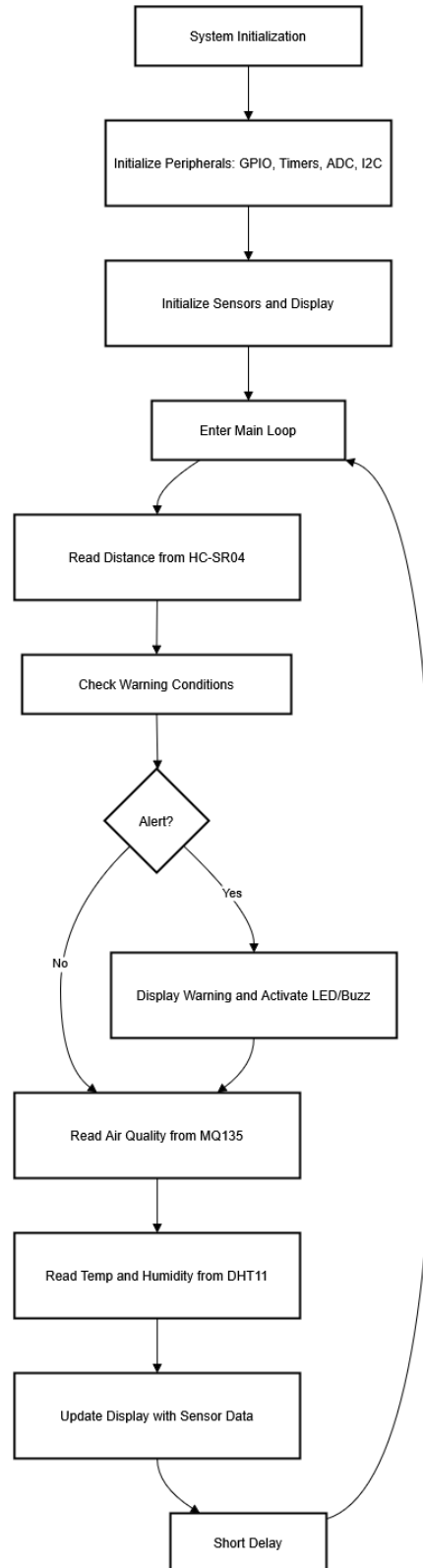
D. Block Diagram/Schematic [CLO-3, 50 Marks]



Pin Connectivity:

STM32 Pin	Connected To	Function
PB9	DHT11 Data	Bidirectional data line for DHT11 sensor
PA1	MQ135 Output	Analog input for air quality sensor
PE10	HC-SR04 TRIG	Output trigger signal for ultrasonic sensor
PA0	HC-SR04 ECHO	Input capture for echo signal timing
PD14	Warning LED	Visual warning indicator
PA5	Buzzer	Audio warning output
PB6	SSD1306 SCL	I2C clock line for OLED display
PB7	SSD1306 SDA	I2C data line for OLED display

E. Flow Chart [CLO-3, 50 Marks]



F. CEP (Project Complexity) Attributes [CLO-3, 50 Marks]

WP1: Depth of knowledge

The project incorporates in-depth knowledge of several key microcontroller concepts:

1. **Timer Configuration and Usage:** Implementation of precise timing mechanisms using both basic counting and input capture modes.
 - TIM1 configured for microsecond delays
 - TIM2 configured for input capture with interrupts
2. **Interrupt Handling:** Proper configuration and use of interrupt-driven programming for real-time response.
 - Input capture interrupts for distance measurement
 - Proper interrupt priority management
3. **Sensor Interfacing:** Implementation of multiple communication protocols:
 - Custom bit-banging protocol for DHT11 sensor
 - Analog-to-digital conversion with calibration formulas for MQ135
 - Ultrasonic timing protocol for HC-SR04
4. **Display Interfacing:** I2C communication with the SSD1306 OLED display, including:
 - Proper initialization sequences
 - Graphics rendering functions
 - Text and UI element display
5. **Data Processing Algorithms:** Conversion of raw sensor data to meaningful metrics:
 - Converting ADC readings to air quality index
 - Processing ultrasonic pulse timing to distance
 - Checksum verification for DHT11 readings

WP2: Range of conflicting requirements

The project faced several conflicting requirements that needed resolution:

1. **Limited GPIO Pin Resources:**
 - Required multiple pins for various sensors and outputs
 - DHT11 needs bidirectional pin configuration
 - Solution: Carefully planned pin assignments and shared functionality where possible
2. **Timing Requirements:**

- DHT11 requires precise microsecond-level timing
- HC-SR04 needs accurate echo pulse timing
- Display updates need to be regular but not interfere with sensor reading
- Solution: Utilized two separate timers and interrupt-based handling

3. **Processing Power Constraints:**

- Real-time computation of AQI from MQ135 sensor
- Distance calculation from HC-SR04 echo time
- Running display update routines
- Solution: Optimized code and used interrupt-driven architecture

4. **Power Management:**

- Sensors have different power requirements
- MQ135 needs warmup time for accurate readings
- Display updates consume processing time
- Solution: Balanced sensor reading frequency with display update rate

WP5: Extent of applicable codes

The project required development of code that extended beyond standard libraries:

1. **Custom DHT11 Protocol Implementation:**

- Developed bit-banging implementation for the proprietary DHT11 protocol
- Created precise timing functions for signal detection
- Implemented error checking with checksum verification

2. **Air Quality Calculation Algorithm:**

- Developed custom formulas to convert sensor resistance to ppm
- Created AQI scaling algorithm based on environmental standards
- Implemented status determination based on calculated values

3. **Ultrasonic Distance Measurement:**

- Developed input capture-based echo timing measurement
- Created distance calculation algorithm with speed of sound compensation
- Implemented warning threshold detection

4. **User Interface Implementation:**

- Created display layout with multiple information sections
- Developed progress bar visualization for air quality
- Implemented warning screen logic and transitions

WP7: Interdependence

The project demonstrates high interdependence between components:

1. **Hardware-Software Integration Challenges:**

- DHT11 sensor required precise software timing to match hardware characteristics
 - HC-SR04 needed hardware timer capture synchronized with software processing
 - Solution: Careful integration of hardware peripherals with software timing
- 2. Sensor Data Dependencies:**
- Warning system depends on accurate readings from multiple sensors
 - Display updates require valid data from all sensors
 - Solution: Implemented error checking and fallback mechanisms
- 3. Critical Timing Dependencies:**
- DHT11 communication has strict timing requirements
 - HC-SR04 echo measurement depends on timer precision
 - Solution: Used hardware timers with appropriate resolution
- 4. Resource Sharing Challenges:**
- Multiple peripherals sharing the same bus (I2C)
 - Interrupt priorities needed careful management
 - Solution: Implemented proper resource management and synchronization

G. Code [CLO-3, 50 Marks]

- 1. Peripheral Initialization:**
- Proper configuration of GPIO, Timers, ADC, and I2C
 - Hardware-specific settings for STM32F407VG
- 2. Sensor Interface Implementations:**
- DHT11 bit-banging protocol with timing control
 - MQ135 ADC reading with conversion algorithms
 - HC-SR04 trigger/echo handling with timer capture
- 3. Display Management:**
- SSD1306 initialization and control
 - Dynamic UI updates with sensor readings
 - Warning screen implementation
- 4. Interrupt Handlers:**
- Input capture interrupt for HC-SR04 echo timing
 - Proper interrupt configuration and priority management

H. References

1. STM32F4 Reference Manual:
https://www.st.com/resource/en/reference_manual/dm00031020-stm32f405-415-stm32f407-417-stm32f427-437-and-stm32f429-439-advanced-arm-based-32-bit-mcus-stmicroelectronics.pdf
2. DHT11 Sensor Implementation Guide: <https://controllerstech.com/dht11-sensor-with-stm32/>
3. HC-SR04 Ultrasonic Sensor with STM32: <https://controllerstech.com/hcsr04-ultrasonic-sensor-with-stm32/>
4. MQ135 Gas Sensor Library: <https://github.com/GeorgK/MQ135>
5. SSD1306 OLED Library for STM32: <https://github.com/afiskon/stm32-ssd1306>
6. STM32 ADC Programming Guide: <https://controllerstech.com/stm32-adc-single-channel/>
7. STM32 Timer Input Capture Tutorial: <https://controllerstech.com/timer-input-capture-in-stm32/>
8. Air Quality Index (AQI) Calculation Methods:
<https://www.airnow.gov/sites/default/files/2020-05/aqi-technical-assistance-document-sept2018.pdf>
9. https://www.micropeta.com/video40#google_vignette
10. <https://github.com/afiskon/stm32-ssd1306>